

Tennis Data Analysis Dashboard.

1. Problem Statement:

The tennis ecosystem generates vast amounts of data across competitions, players, rankings, and venues — yet this data remains scattered, unstructured, and inaccessible for meaningful analysis. Sports analysts and organizations struggle to efficiently manage and interpret this data from the SportRadar API, as the raw JSON format lacks structure, making it difficult to explore tournament hierarchies, analyze competition trends, and extract event-specific insights. The absence of a centralized system that organizes and presents this data effectively leaves analysts, fans, and organizations without the tools needed to make informed, data-driven decisions.

The final dashboard is live and accessible at <https://tennis-project-analysis.streamlit.app/>

2.Data Source :

All data was collected from the **SportRadar Tennis API**, a professional sports data provider delivering real-time tennis data via RESTful endpoints in JSON format. Authentication is handled through an API key obtained from the SportRadar Developer Console.

3. Data Extraction

- Authenticated with the **SportRadar Tennis API** using a unique **API key**.
- API endpoints were called using Python's **requests** library to retrieve raw data.
- Data was received in **JSON format** — nested and unstructured.
- Parsed the JSON response and flattened nested fields into a **structured tabular format** using **Pandas**.
- Saved each dataset as a **CSV file** for cleaning and further processing.

4. Data Cleaning & Transformation

- Identified missing values in critical columns and handled them through imputation.
- Removed duplicates across all datasets using `drop_duplicates()`.
- Stripped SportRadar prefixes from all ID fields to extract clean numeric identifiers.
- Renamed columns to align with the target PostgreSQL schema.
- Optimized data types — converted categorical fields to category dtype.
- Exported cleaned datasets as CSV files, ready for database ingestion.

5. Database & Schema Creation

5.1 Database Creation

- Database was created using **Neon**, a cloud-based PostgreSQL service.
- Different **user roles** were defined and assigned appropriate permissions to manage access control among team members.
- Each team member was provided with database credentials — host, port, username, password, and database name.
- Database was connected to **pgAdmin** for query execution, table management, and data interaction.

5.2 Schema Creation

5.2.1 Schema Design

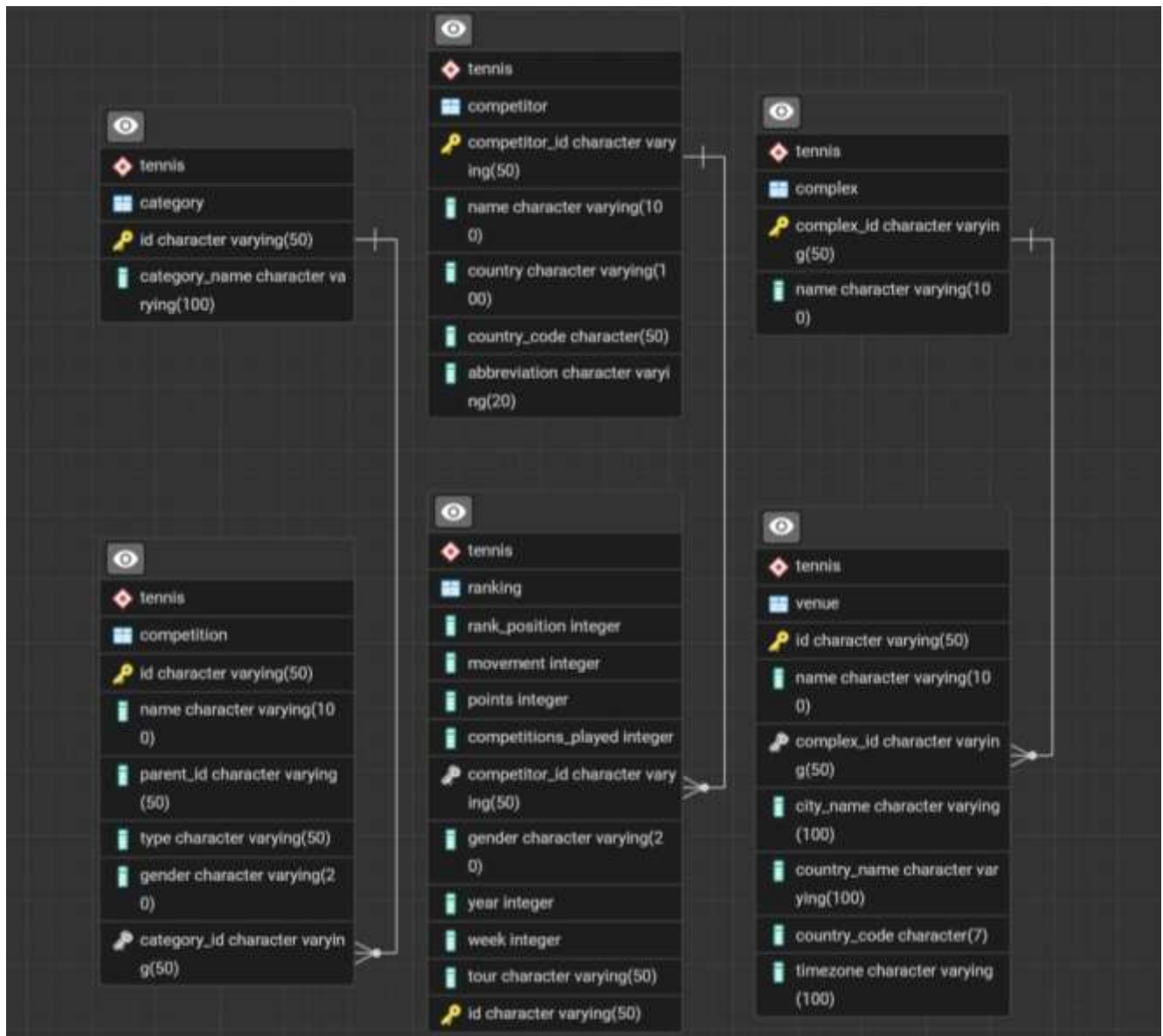
- Designed a normalized schema to eliminate redundancy and ensure referential integrity.
- Defined appropriate **primary keys**, **foreign keys**, and **data types** for each table based on the ERD.
- All ID fields defined as VARCHAR(50) to maintain consistency with SportRadar identifier format.

5.2.2 Table Creation

- category — stores competition categories
- competition — stores all competitions with parent-child hierarchy
- complex — stores sports complexes
- venue — stores venue details linked to complexes
- competitor — stores player information
- ranking — stores player ranking and performance metrics

5.2.3 Entity Relationship Diagram (ERD)

The ERD below represents the complete database structure with all tables, columns, data types, and relationships.



The ERD consists of 6 tables with the following relationships:

- category → competition — One-to-Many via category_id
- competitor → ranking — One-to-Many via competitor_id
- complex → venue — One-to-Many via complex_id
- competition → competition — Self-referencing via parent_id to represent tournament hierarchy

6. Exploratory Data Analysis (EDA)

EDA was performed using SQL queries executed on the **Neon PostgreSQL** database. Queries were divided into two categories — **Analytical** and **Statistical** — to derive meaningful insights from the tennis dataset.

6.1 Analytical Queries

1. How does a player's volume of participation influence their average ranking points, and what is the distribution across participation tiers?
2. Which countries produce the highest-performing tennis players on average?
3. Who are the top 20 players currently active based on ranking points, and which tour do they dominate?
4. Which players demonstrate the highest point efficiency by maximizing ranking gains with minimum tournaments played?
5. How does a player's frequency of competition correlate with their ranking success and rank stability?

6.2 Statistical Queries

1. Which countries have the highest number of professional tennis players represented?
2. What is the distribution of tournament formats (Singles vs. Doubles) across different genders?
3. What are the average ranking points and participation levels for players on ATP vs. WTA tours?
4. How much did world rankings fluctuate this week in terms of average movement, highest climb, and deepest drop?
5. Which countries produce players with the highest average ranking points?

7. Dashboard & Design

The dashboard was developed using **Streamlit** with **5 analytical pages**, each designed to answer specific business questions from the tennis dataset.

7.1 Dashboard Pages

Page 1 — Overview

- Displays key metrics — Total Competitors, Total Competitions, Total Venues, and Total Countries — to provide an immediate snapshot of the dataset.
- Visualizes the distribution of ranking points to understand how points are spread across the competitor base.
- Analyses rank movement distribution to identify how many players improved, declined, or remained stable.

Page 2 — Player Performance

- Examines average competitions played per competitor to understand participation patterns across the player base.
- Investigates the correlation between competitions played and ranking points to determine whether higher participation leads to better rankings.
- Segments competitors into rank tiers (Top 10, 11–50, 51–100, 100+) to compare performance and participation levels across competitive levels.

Page 3 — Ranking & Competition Dynamics

- Identifies rank movement outliers — top 5% gainers and losers — to understand what drives extreme ranking changes.
- Analyzes competition category hierarchy to determine which categories contain the most sub-competitions.

Page 4 — Country Analysis

- Identifies countries producing the most top-ranked competitors (top 50) to highlight dominant tennis nations.
- Calculates a country dominance index — the percentage of elite competitors (top 50) relative to total competitors per country.
- Examines geographic mismatch between player origin and venue location to identify underserved markets.

Page 5 — Infrastructure

- Analyses venue distribution across countries to identify which nations have the strongest tennis infrastructure.
- Evaluates complex-level concentration to understand where multi-venue facilities are clustered globally.
- Assesses competition hierarchy completeness by identifying orphaned competitions per category to flag data quality issues.

7.2 Design

- Refined the overall layout to ensure a clean, structured, and readable presentation across all pages.
- Implemented **custom CSS** to control colours, spacing, and chart alignment for a consistent visual experience.
- Applied a **dark theme** to enhance visual clarity and deliver a modern, professional look.
- Configured **.streamlit/config.toml** to define primary colours, background settings, and accent styles.
- Maintained a **consistent colour palette** across all pages to ensure visual coherence.
- Optimized chart sizes and placements to maintain a balanced and focused layout throughout.
- Final design follows **modern BI dashboard principles** — clean, intuitive, and accessible for both technical and non-technical users.
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8. Insights

- Rank volatility dominates — more players lost positions than gained, proving consistency beats participation.
- ITF accounts for 65%+ of all competitions — the sport's foundation is grassroots, not elite circuits.
- Near-zero correlation ($r=0.009$) between competitions played and points — performance quality drives rankings, not volume.
- Top 10 players earn 7x more points than the 51–100 tier — ranking wealth is heavily concentrated at the elite level.
- USA leads with 103 competitors, nearly double second-placed France — reflecting superior national development infrastructure.
- Small nations like Denmark and Norway convert a higher percentage of players into top-50 — quality of development outweighs pool size.
- Top rank gainers played moderate schedules — strategic tournament selection drives the biggest rank jumps.
- Top rank losers played heavy schedules — overexposure and fatigue are the primary drivers of ranking decline.
- USA hosts 3x more venues than any other country, yet many top-50 players come from countries with zero venues — a clear infrastructure-talent mismatch.
- ITF Men and Women have the most orphaned competitions — data incompleteness is a grassroots problem, not an elite one.

9. Business Growth Plan

For Tennis Governing Bodies (ATP, WTA, ITF)

- ITF represents 65%+ of all competitions — targeted investment in ITF infrastructure directly strengthens the future ATP/WTA talent pipeline.
- Missing data concentrated in ITF categories highlights a data governance gap that needs to be addressed to ensure accurate global rankings.
- High elite conversion rates in smaller nations like Denmark and Norway offer a replicable blueprint for efficient national player development.

For Sponsors and Broadcasters

- Ranking points are heavily concentrated in the top 10 — sponsorship targeting should align with where visibility and returns are highest.
- Players with strategic scheduling and sharp rank movement represent the best early-stage sponsorship opportunities before market value peaks.
- Points-per-competition efficiency is a more reliable indicator of player commercial value than total matches played.

For Event Organizers

- Heavy dependence on USA venue infrastructure creates geographic risk — diversifying tournament locations is essential for sustainable global tour growth.
- South America is a significantly underleveraged market — Buenos Aires alone hosts 29 venues, signaling strong readiness for premium events.
- Countries producing top-50 players but hosting no venues are the most viable targets for new tournament development.

For Sports Analysts

- Top rank losers consistently played heavy schedules — competition load monitoring is a reliable early warning signal for ranking decline.
- USA, Czechia and France produce top-50 players but host disproportionately fewer venues — this mismatch is a measurable market opportunity.
- ITF orphaned data is a known quality issue — any predictive model built on competition hierarchy data must account for this gap.

10. Conclusion

This project delivered a complete end-to-end data analytics pipeline — from SportRadar API extraction to an interactive Streamlit dashboard. Data across 1,000 competitors, 6,477 competitions, and 3,891 venues was extracted, cleaned, stored in a normalized PostgreSQL database, and visualized across 5 analytical pages.

The analysis revealed that ranking success is driven by performance quality over volume, elite talent is geographically concentrated, and significant infrastructure-talent mismatches exist globally — all presenting measurable opportunities for stakeholders across the sport.

11. Future Improvements:

- Automate weekly data refresh from SportRadar API.
- Expand to singles rankings for complete player analysis.
- Integrate match-level data for win/loss and head-to-head analysis.
- Build a predictive model for rank movement forecasting based on schedule load.

12. Technologies Used

- **Python** — Primary programming language used for data extraction, cleaning, transformation, and database loading.
- **VS Code** — Code editor used for writing and managing all scripts and application code.
- **PostgreSQL (Neon)** — Cloud-based relational database used for structured data storage and SQL query execution.
- **Streamlit** — Used to build the interactive web dashboard for data exploration and visualization.
- **Git & GitHub** — Used for version control and project code management.